3Consulting
BUILD TRUST IN THE DIGITAL ECONOMY

REAL-TIME FRAUD PREVENTION · CASE STUDY

FraudGuard: Sub-50ms Transaction Risk & SAR Queue

BI Scientist • AI/ML Engineer • Python Developer

Enterprise fraud scoring engine combining webhook ingestion, real-time feature stores, 0–1000 risk scoring, and automated compliance alert queues.

PAYMENT VULNERABILITIES

The Speed & Precision Dilemma in Modern Payments

Fintech payment rails process thousands of requests per second where a 100ms latency delay causes drop-offs, but undetected fraud destroys unit economics.

Account Takeover (ATO)

- Credential stuffing and SIM-swap fraud breach accounts in seconds.
- Sudden IP, device fingerprint, and geographic jumps bypass static rules.
- Need dynamic velocity counters across 1-min to 24-hr windows.

Card Velocity Bursts

- Fraud rings test stolen BIN ranges with micro-charges.
- Standard daily volume limits fail to detect rapid 10-second bursts.
- Results in massive chargeback fees and merchant processor bans.

Analyst Fatigue & SARs

- Traditional systems flood analysts with 90% false-positive alerts.
- Legitimate payments get blocked while real fraud slips through.
- Regulatory Suspicious Activity Reports (SARs) are delayed.

SYSTEM TOPOLOGY

High-Throughput Redis Buffers & Dynamic Risk Scoring

Engineered for sub-50ms decisioning with clear decision reason codes.

Webhook Ingestion

- Asynchronous payload verification.
- Zero blocking of checkout authorization.
- Fast buffer queue via Redis Streams.

Dynamic Risk Engine

- 0–1000 normalized risk score.
- Top-3 reason codes attached per alert.
- Automated allow/challenge/block rules.

Analyst Workbench

- Real-time SAR prioritization queue.
- One-click dispute evidence packaging.
- Human-in-the-loop decision feedback.

ANALYST WORKBENCH

FraudGuard Operations Console

Real-time payment risk scoring console showing live transactions and reason codes.

- **Instant Decisioning:** Green (<300), Yellow (300–700), Red (>700) scoring bands for instant merchant gateway routing.
- **Top Reason Codes:** Explains exactly why a transaction was flagged (e.g. `VELOCITY_1H_EXCEEDED`, `NEW_DEVICE_HIGH_VAL`).
- **Integrated SAR Workflow:** Escalates confirmed fraud directly to Nigerian Financial Intelligence Unit (NFIU) reporting standards.

FastAPI | Redis | PostgreSQL | SAR Workflow



fraudguard.3consult-ng.com/app/queue

QUANTIFIED RESULTS

Performance at High Velocity

Production benchmarks proving accuracy and sub-second execution.

Scoring Latency	Analyst Load	False Positives	Reason Explainability
< 50 ms	-65%	Down 48%	100%
Sub-100ms merchant SLA	Prioritized high-risk queue	Calibrated probability cutoff	Top-3 codes on every alert

Operational Efficiency

- Analysts focus exclusively on high-risk, high-impact suspicious events.
- Automated rule challenge reduces customer friction on genuine payments.
- End-to-end audit trail for merchant bank dispute chargebacks.

Scalable Infrastructure

- High-concurrency async ASGI workers handling payment bursts.
- Decoupled scoring from compliance archiving via background workers.
- Zero downtime during rule threshold updates.

CASE STUDY TAKEAWAY

Real-time risk scoring without checkout friction.

FraudGuard proves that fraud detection doesn't have to slow down checkout or swamp compliance teams with false alerts.

Company: 3Consulting Limited (Lagos, Nigeria)
Live Platform: fraudguard.3consult-ng.com